- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

15 A projectile is launched from a point on level ground with an initial velocity u at an angle θ above the horizontal.

(a) Show that the range of the projectile is given by $\dfrac{2u^2 \sin\theta\cos\theta}{g}$. **[3]**

(b) Determine the set of values of θ for which the maximum height of the projectile is greater than the range, where θ is an acute angle. Give your answer in degrees. **[5]**

15(a)

15(b)

(answer space continued on next page)

15(b) **(continued)**